



Kimia Farma X Rakamin

KIMIA FARMA PERFORMANCE ANALYSIS

Business Year 2020 - 2023



Introduction

PT. Kimia Farma Tbk.

Kimia Farma was the first pharmaceutical industry company in Indonesia established by the Dutch Indies Government in 1817. The name of this company was originally NV Chemicalien Handle Rathkamp & Co. Based on the policy of nationalization of ex Dutch companies at the beginning of the independence period; the Government of the Republic of Indonesia merged in 1958 a number of pharmaceutical companies into PNF Bhinneka Kimia Farma. Then, on 16 August 1971, the form of PNF as legal entity was changed to Limited Liability Company and the company's name was changed to PT Kimia Farma (Persero).

source: Kimia Farma Official Website

Challenges Given

- Import Dataset to BigQuery
- Create Analysis Table with BigQuery
- Create Dashboard Performance Analysis Business Year 2020-2023



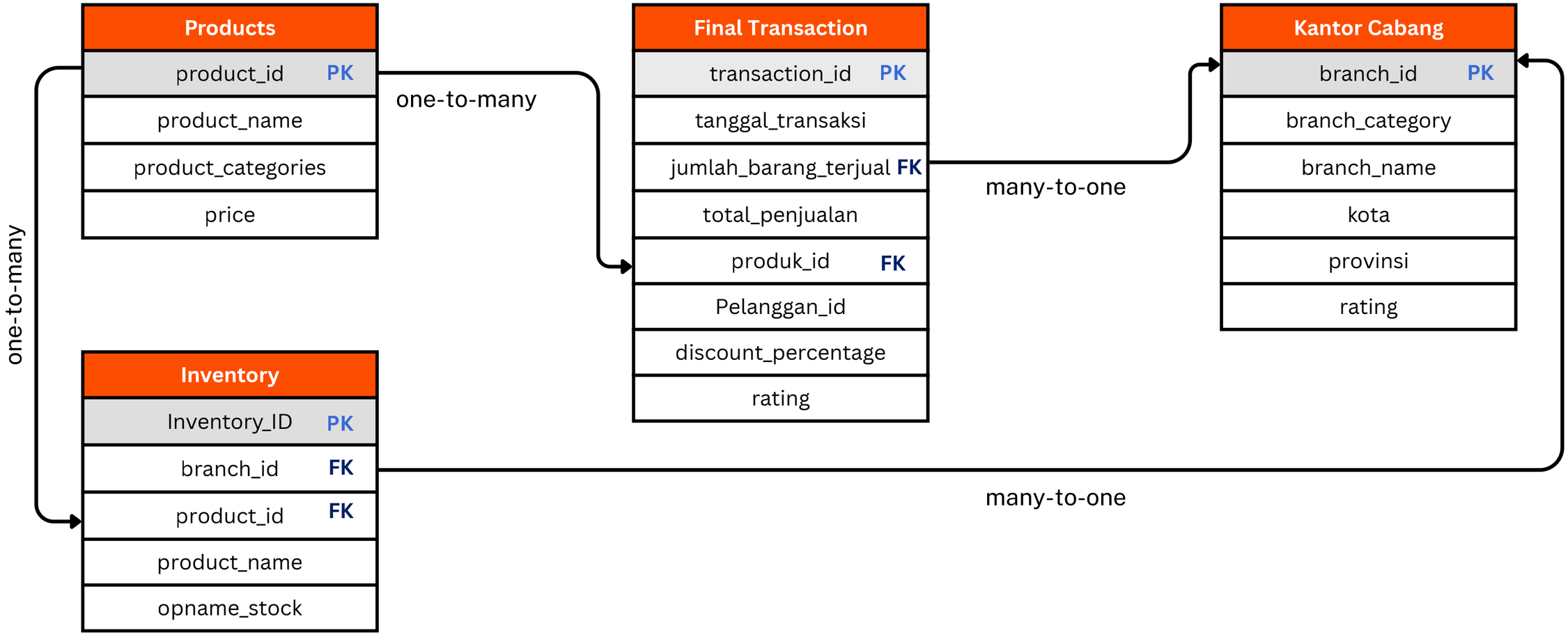
ABOUT ME.

A Career Shifter with hands-on experience in Mortgage Product teams and a strong understanding of customer lending and end-to-end banking business processes. Experienced in collaborating with cross-functional stakeholders to translate business needs into structured requirements and solutions. Currently transitioning into an Analyst role by joining Data Science programs by Rakamin and Digitalent to developed practical skills in Python and SQL to process, analyze, and visualize large datasets, uncover patterns, and derive meaningful insights.

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GitHub : https://github.com/mercyclaudiam-netizen/

Relationship Scheme: Primary Key

This dataset is derived from the PT Kimia Farma business performance, covering the period from **January 2020 to December 2023.**
The **data consists of four core tables** and is structured using **a star-schema design**, with the Final Tarnscation table serving as the central fact table that captures transactional sales activity.



Pre-Processing

The data from each table shown no indication of duplicated data, rows with null value, wrong format or unusual values.

```
df_ft.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 672458 entries, 0 to 672457
Data columns (total 8 columns):
 #   Column              Non-Null Count   Dtype
---  ------              --------------   -----
 0   transaction_id      672458 non-null  object
 1   date                672458 non-null  object
 2   branch_id           672458 non-null  int64
 3   customer_name       672458 non-null  object
 4   product_id          672458 non-null  object
 5   price               672458 non-null  int64
 6   discount_percentage 672458 non-null  float64
 7   rating              672458 non-null  float64
dtypes: float64(2), int64(2), object(4)
memory usage: 41.0+ MB
```

```
df_i.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1035000 entries, 0 to 1034999
Data columns (total 5 columns):
 #   Column        Non-Null Count    Dtype
---  ------        --------------    -----
 0   Inventory_ID  1035000 non-null  object
 1   branch_id     1035000 non-null  int64
 2   product_id    1035000 non-null  object
 3   product_name  1035000 non-null  object
 4   opname_stock  1035000 non-null  int64
dtypes: int64(2), object(3)
memory usage: 39.5+ MB
```

```
df_kc.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1725 entries, 0 to 1724
Data columns (total 6 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   branch_id        1725 non-null   int64
 1   branch_category  1725 non-null   object
 2   branch_name      1725 non-null   object
 3   kota             1725 non-null   object
 4   provinsi         1725 non-null   object
 5   rating           1725 non-null   float64
dtypes: float64(1), int64(1), object(4)
memory usage: 81.0+ KB
```

```
df_p.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 4 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   product_id        150 non-null    object
 1   product_name      150 non-null    object
 2   product_category  150 non-null    object
 3   price             150 non-null    int64
dtypes: int64(1), object(3)
memory usage: 4.8+ KB
```

Row	total_rows	null_product_id	null_product_name	null_product_cate...	null_price
1	150	0	0	0	0

Row	total_rows	null_inventory_id	null_branch_id	null_product_id	null_product_name	null_opname_stock
1	1035000	0	0	0	0	0

Row	total_row	null_branch_id	null_branch_cate...	null_branch_name	null_kota	null_provc	null_rating
1	1725	0	0	0	0	0	0

Row	total_row	null_transaction_id	null_date	null_branch_id	null_customer_na...	null_product_id	null_price	null_discount_per...	null_rating
1	672458	0	0	0	0	0	0	0	0

Master Table

Data from the **four source tables** were **integrated using SQL to create a consolidated Master Table** to help analytic process.
This **Master Table summarizes key sales information** for PT Kimia Farma and serves as a single, analytics-ready source, enabling more efficient dashboard development and faster insight generation.

```sql
WITH sales_order as (
    SELECT
      ft.transaction_id, ft.date, ft.branch_id, ft.customer_name, ft.product_id, ft.price as actual_price, ft.discount_percentage, ft.rating as rating_transaksi,
      (CASE
            WHEN ft.price <= 50000 THEN 0.1
            WHEN ft.price <= 100000 THEN 0.15
            WHEN ft.price <= 300000 THEN 0.2
            WHEN ft.price <= 500000 THEN 0.25
            ELSE 0.3 END ) AS persentase_gross_laba,
      ft.price-(ft.price*ft.discount_percentage) AS nett_sales
    FROM `rakamin-pbi-kimia-farma-490322.kimia_farma.final_transaction` ft)
SELECT
  so.transaction_id,so.date,so.branch_id,kb.branch_name,kb.kota,kb.provinsi,kb.rating as rating_cabang,so.customer_name,so.product_id,p.product_name,so.
actual_price,so.discount_percentage,so.persentase_gross_laba,so.nett_sales,so.nett_sales * so.persentase_gross_laba as nett_profitt,so.rating_transaksi
FROM sales_order so
JOIN `rakamin-pbi-kimia-farma-490322.kimia_farma.kantor_cabang` kb
ON so.branch_id = kb.branch_id
JOIN `rakamin-pbi-kimia-farma-490322.kimia_farma.product` p
ON so.product_id = p.product_id
GROUP BY so.date,so.transaction_id,so.date,so.branch_id,kb.branch_name,kb.kota,kb.provinsi,kb.rating,so.customer_name,so.product_id,p.product_name,so.
actual_price, so.discount_percentage,so.persentase_gross_laba,so.nett_sales,so.rating_transaksi
ORDER BY so.date ASC;
```

This script will process 206.47 MB when run.

Row	transaction_id	date	branch_id	branch_name	kota	provinsi	rating_cabang	customer_name	product_id	product_name	actual_price	discount_percent...	persentase_gros...	nett_sales	nett_profitt	rating_transaksi
1	TRX3248504	2020-01-01	92582	Kimia Farma - Apotek	Sibolga	Sumatera Utara	4.4	Susan Stevens DDS	KF162	Anti-inflammatory and antirheumatic products, non-steroids, Propionic acid derivatives	840200	0.01	0.3	831798.0	249539.4	3.0
2	TRX9756282	2020-01-01	43663	Kimia Farma - Klinik-Apotek-Lab...	Ambon	Maluku	4.0	Jeffrey Lutz	KF162	Anti-inflammatory and antirheumatic products, non-steroids, Propionic acid derivatives	840200	0.0	0.3	840200.0	252060.0	3.0
3	TRX2604879	2020-01-01	48243	Kimia Farma - Klinik & Apotek	Tasikmalaya	Jawa Barat	4.1	Jay Mitchell	KF223	Anti-inflammatory and antirheumatic products, non-steroids, Acetic acid derivatives and related substances	401700	0.15	0.25	341445.0	85361.25	3.0

Link to BigQuery

Pre-Processing After Join

Collab

```
df_rs.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 672458 entries, 0 to 672457
Data columns (total 16 columns):
 #   Column                 Non-Null Count   Dtype  
---  ------                 --------------   -----  
 0   transaction_id         672458 non-null  object 
 1   date                   672458 non-null  object 
 2   branch_id              672458 non-null  int64  
 3   branch_name            672458 non-null  object 
 4   kota                   672458 non-null  object 
 5   provinsi               672458 non-null  object 
 6   rating_cabang          672458 non-null  float64
 7   customer_name          672458 non-null  object 
 8   product_id             672458 non-null  object 
 9   product_name           672458 non-null  object 
 10  actual_price           672458 non-null  int64  
 11  discount_percentage    672458 non-null  float64
 12  persentase_gross_laba  672458 non-null  float64
 13  nett_sales             672458 non-null  int64  
 14  nett_profitt           672458 non-null  float64
 15  rating_transaksi       672458 non-null  float64
dtypes: float64(5), int64(3), object(8)
memory usage: 82.1+ MB
```

```
df_rs.duplicated().sum()
```

```
np.int64(0)
```

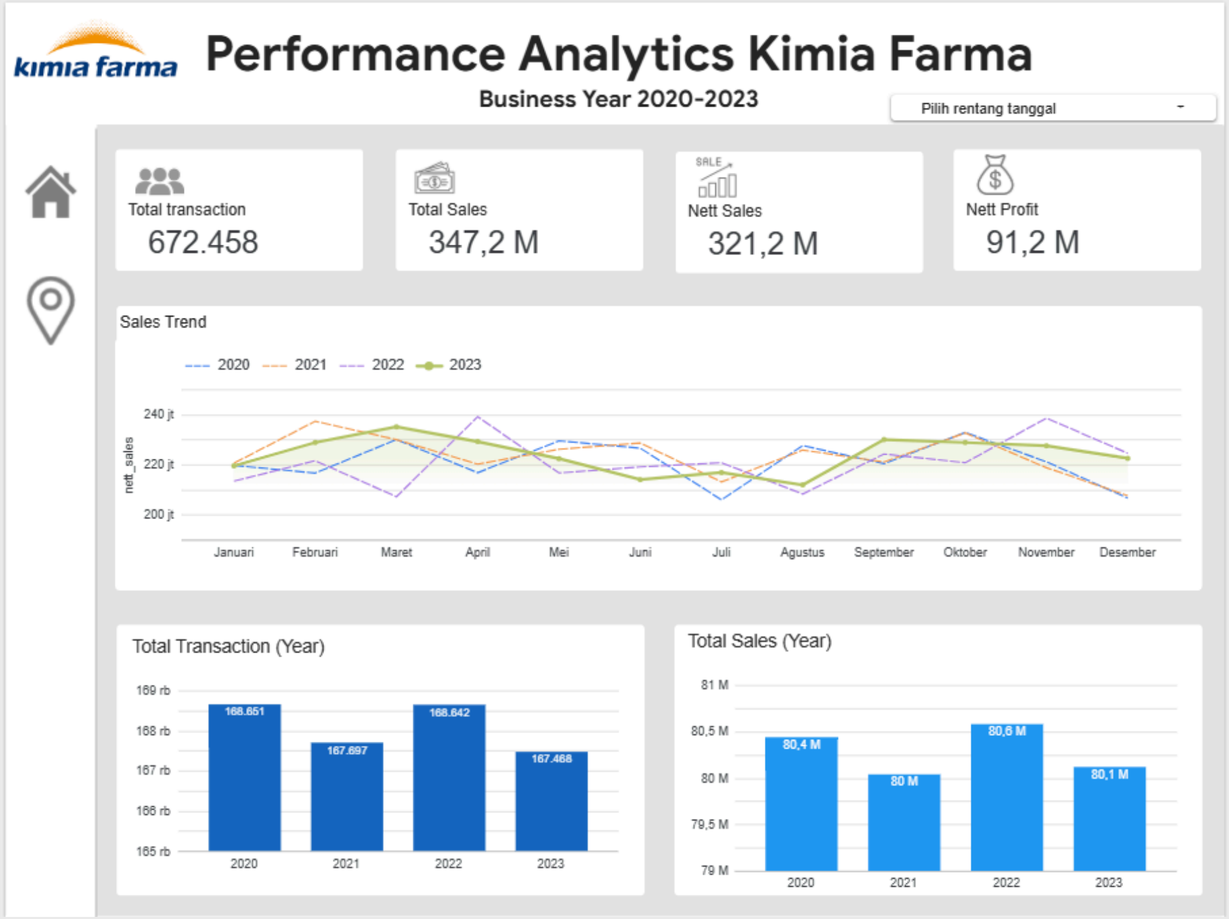
```
df_rs.describe()
```

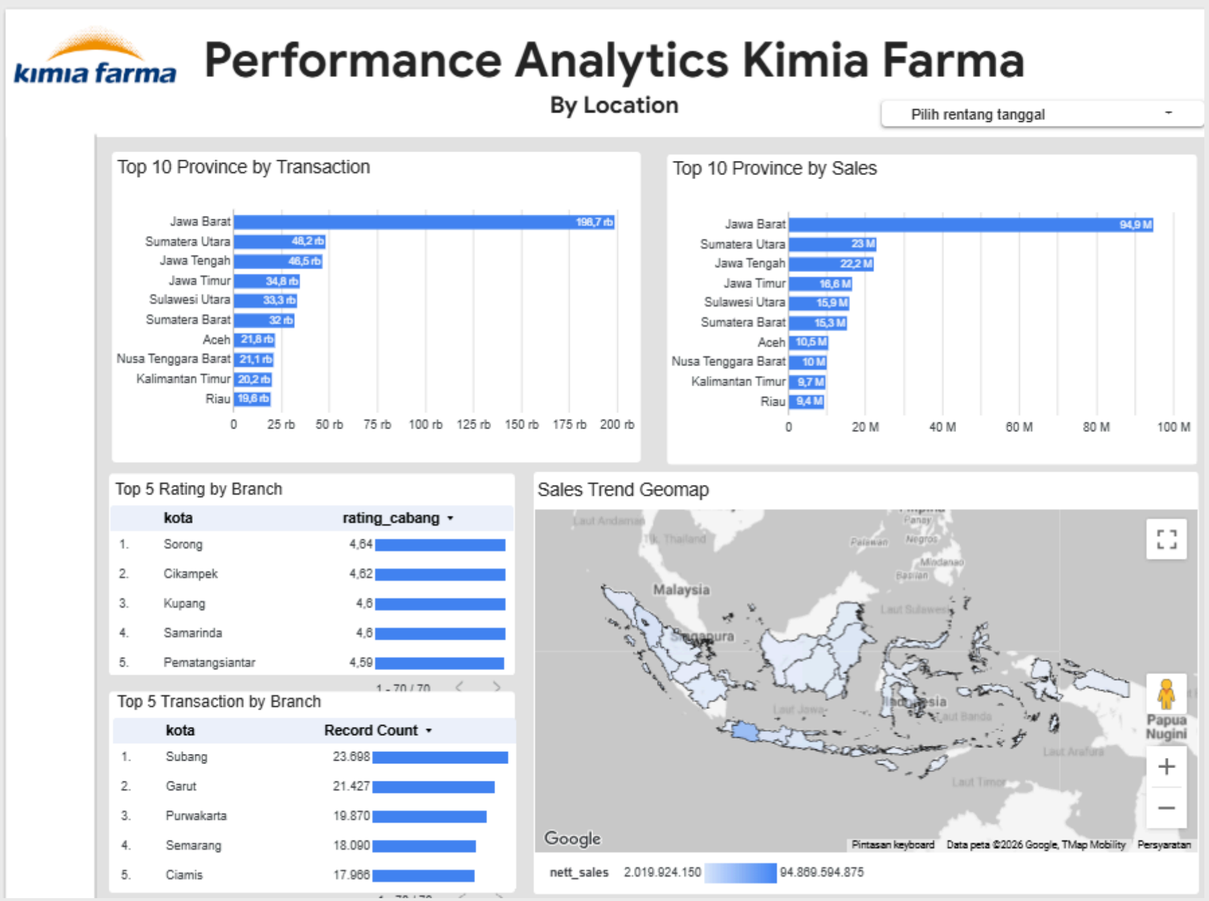
	branch_id	rating_cabang	actual_price	discount_percentage	persentase_gross_laba	nett_sales	nett_profitt	rating_transaksi
count	672458.000000	672458.000000	672458.000000	672458.000000	672458.000000	672458.000000	672458.000000	672458.000000
mean	54232.575093	4.447984	516347.954519	0.074992	0.256225	477607.806464	135644.141433	4.000033
std	26025.980201	0.323350	284547.593398	0.043477	0.056106	264422.423296	85148.115522	0.579164
min	11079.000000	3.900000	2100.000000	0.000000	0.100000	1785.000000	178.500000	3.000000
25%	31016.000000	4.200000	296900.000000	0.040000	0.200000	273702.000000	56411.000000	3.500000
50%	53629.000000	4.500000	523100.000000	0.070000	0.300000	480150.000000	142444.500000	4.000000
75%	76442.000000	4.700000	757900.000000	0.110000	0.300000	703380.000000	211014.000000	4.500000
max	99997.000000	5.000000	997500.000000	0.150000	0.300000	997500.000000	299250.000000	5.000000

To make sure that the data that will be visualized for performance analysis is clean, the result of consolidating tables from dataset need to be checked to find missing value, duplicated rows, wrong format, and unusual data.

Using Python, the data was found to be clean. with no null values, duplicated, wrong format and unusual data.

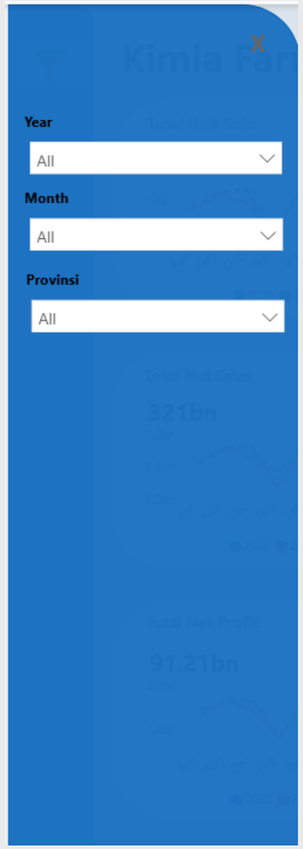
Looker Studio



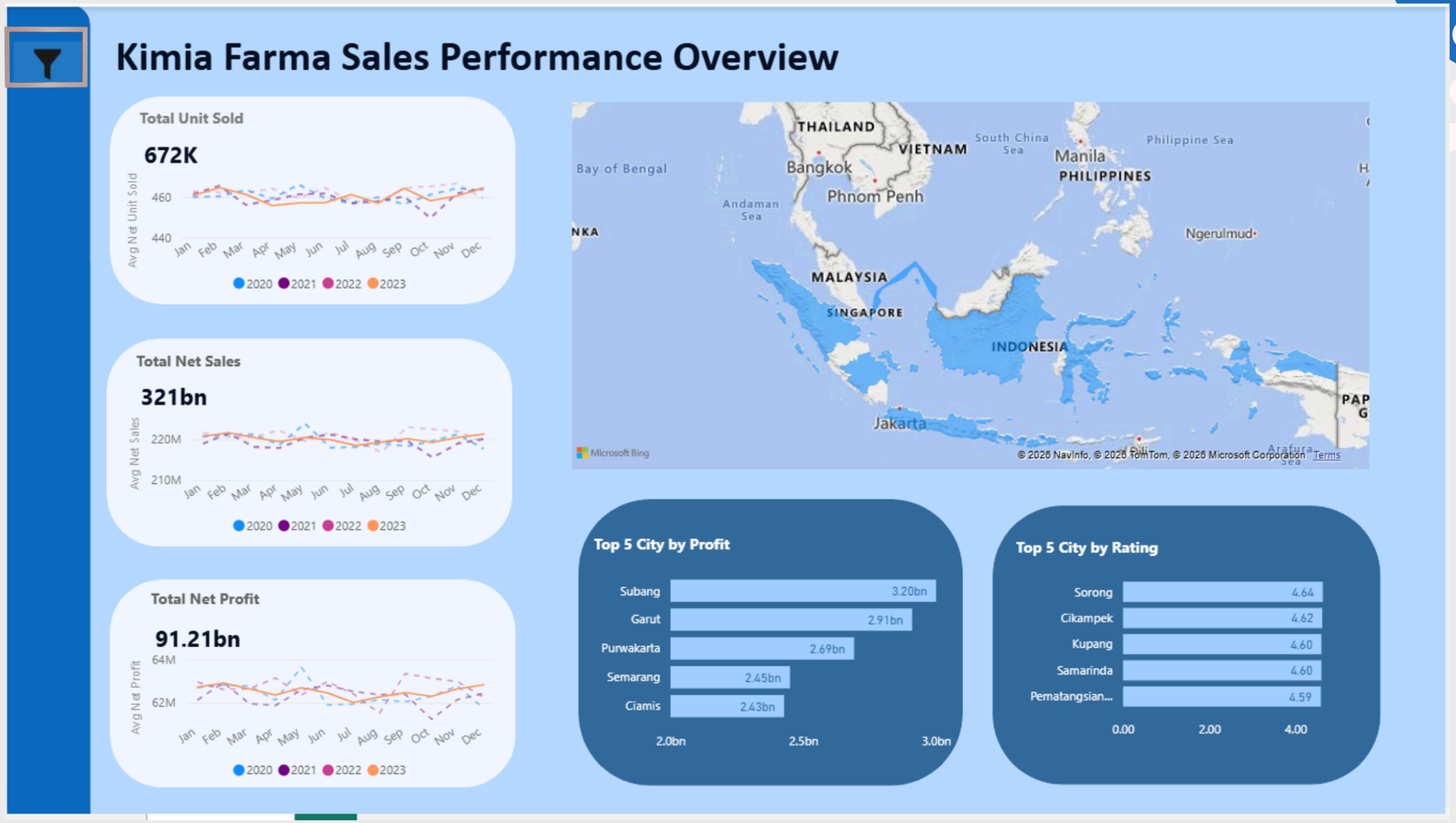


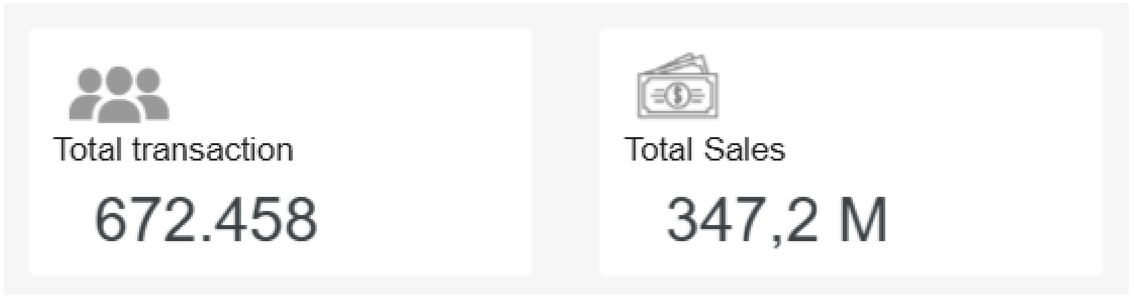
Dashboard Result

Power BI



Click the filter button to show up the pop up filter tab





Dashboard Analysis & Insight

Sales Frequency

Average transaction about +/- 478k per transaction. Quite big number with frequent transaction.

Margin Rate

The company generated 321.2M net sales from 672k transactions, resulting in 91.2M net profit with a healthy 28% margin

Discounts Effect

Discounts reduced revenue by 26M (7.5%), but overall profitability remains strong.

Link to Dashboard



source: Kimia Farma looker dashboard

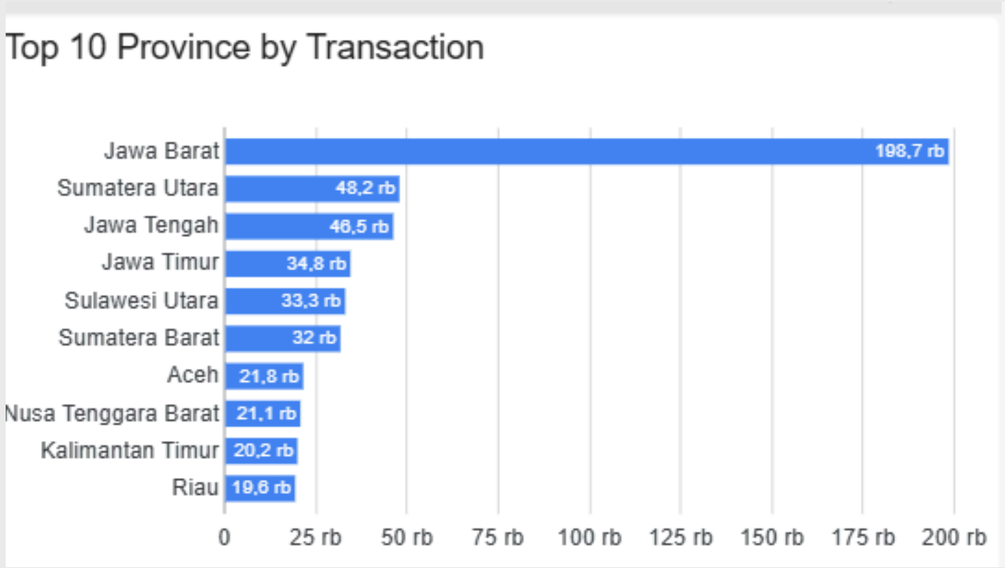


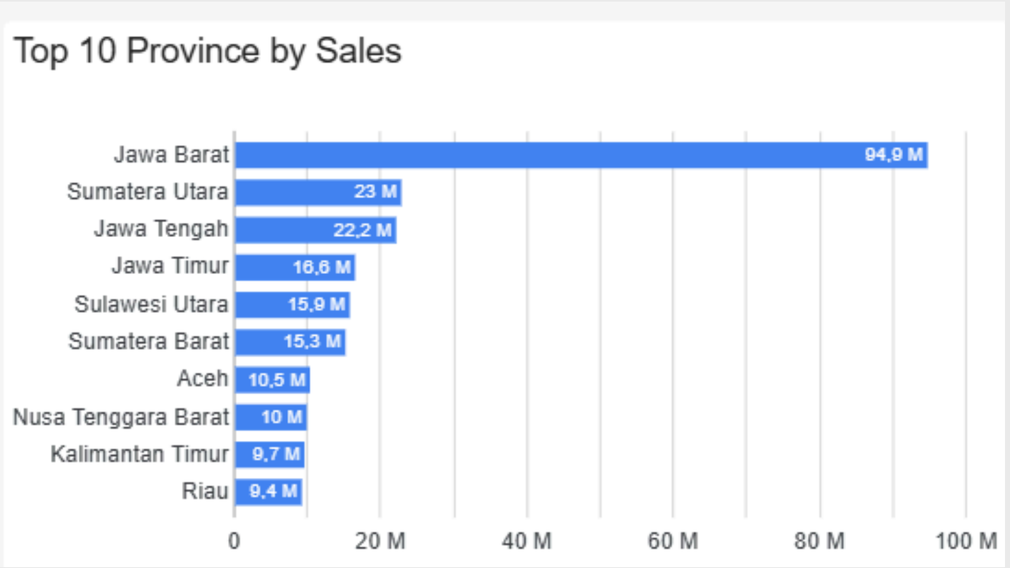
source: Kimia Farma Power BI Dashboard

Dashboard Analysis

- Net sales are stable at about Rp80bn per year. Monthly net sales stay flat (about Rp220M) all year round.
- Sales Performance & Transaction Volume relatively stable.from each year.

Dashboard Analysis





Top sales and transaction is dominated by West Java by a really wide margin (4x from the second most performing region. This may indicate that the company has weak penetration in secondary region. To improve the performance from other region the company needs to analyze the market segment & strategy in West Java and also benchmarking with competitors in secondary area.

Top 5 Rating by Branch

	kota	rating_cabang
1.	Sorong	4,64
2.	Cikampek	4,62
3.	Kupang	4,6
4.	Samarinda	4,6
5.	Pematangsiantar	4,59

1 - 70 / 70

Lowest 5 Rating by Branch

	kota	rating_cabang
1.	Magelang	4,25
2.	Banda Aceh	4,26
3.	Padang	4,3
4.	Lubuklinggau	4,31
5.	Bontang	4,32

1 - 70 / 70

Top 5 Transaction by Branch

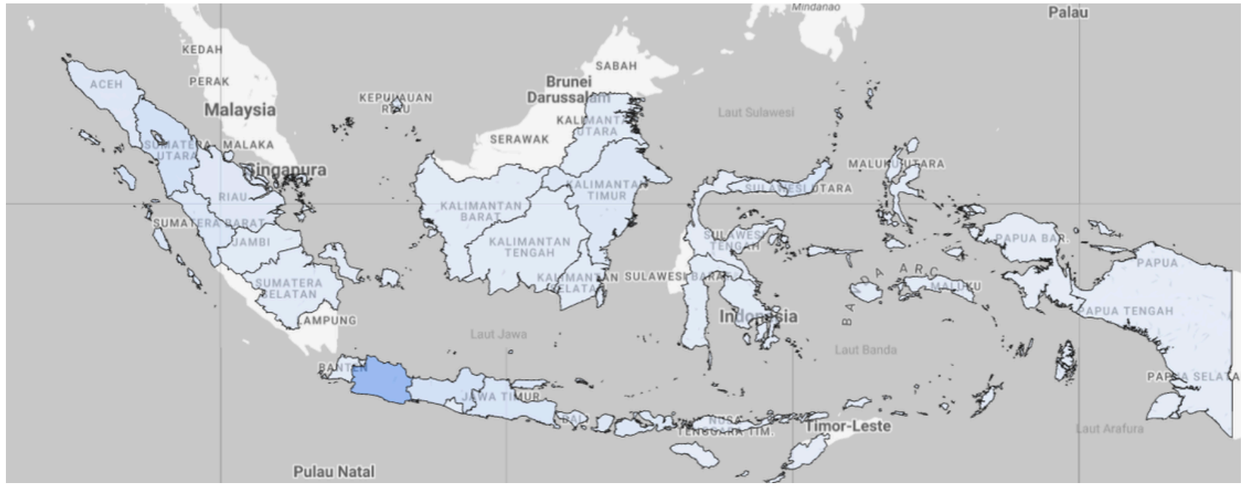
	kota	Record Count
1.	Subang	23.698
2.	Garut	21.427
3.	Purwakarta	19.870
4.	Semarang	18.090
5.	Ciamis	17.966

1 - 70 / 70

Lowest 5 Transaction by Branch

	kota	Record Count
1.	Sorong	4.241
2.	Bontang	4.508
3.	Cianjur	4.932
4.	Palu	5.450
5.	Lubuklinggau	5.800

1 - 70 / 70



1. Although customer satisfaction varies by location, overall service quality appears fairly uniform, as even the lowest-rated branches maintain ratings above 4.20, indicating generally positive customer experiences.
2. Transaction volume does not necessarily align with service ratings. Branches with the highest transaction counts do not appear among the top-rated branches.
3. High transaction branches are concentrated in Java, especially West Java.

THANK YOU